

Exp no: 13 Determination and plotting of intrinsic impedance and phase velocity of various medium

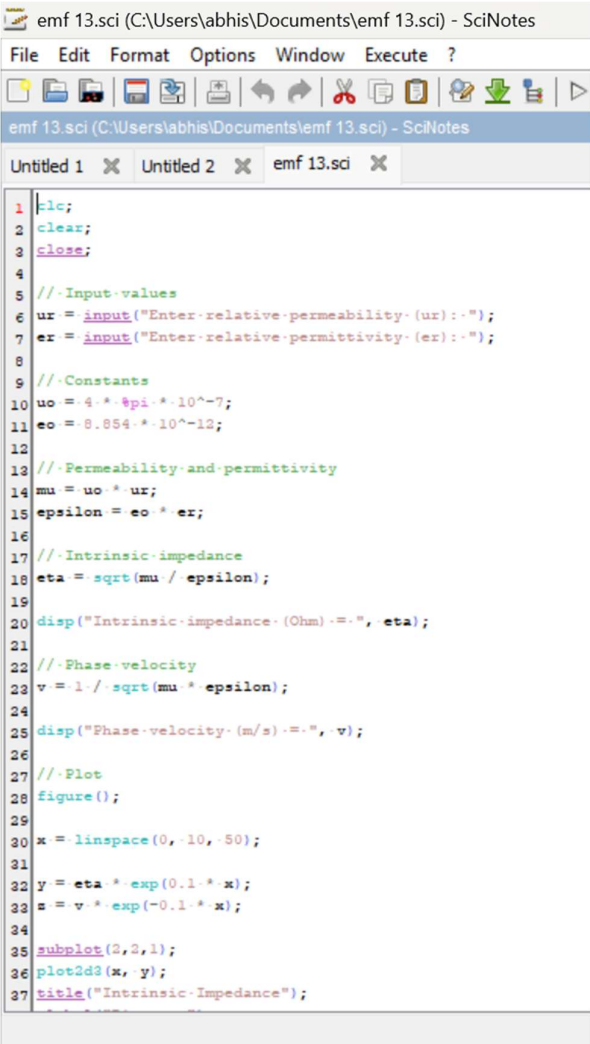
Aim: To determine and plot intrinsic impedance and phase velocity of various medium.

Software required:

- SCILAB version 6.0.1

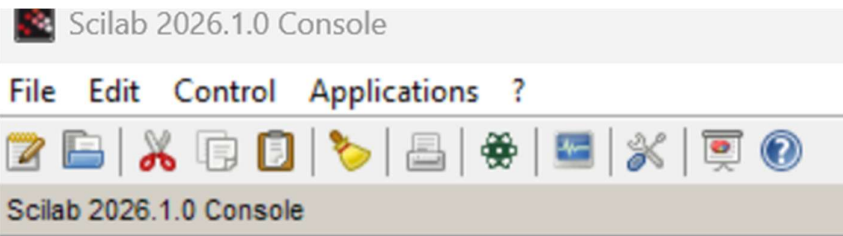
Theory:

Intrinsic Impedance (Z) The intrinsic impedance of a medium is an important parameter that characterizes the relationship between the electric and magnetic fields in an electromagnetic wave propagating through that medium. It represents the opposition that the medium presents to the electric and magnetic fields. $\square$ Definition: Intrinsic impedance Z is the ratio of the electric field (E) to the magnetic field (H) in an electromagnetic wave propagating in that medium. In other words, it describes how much resistance the wave experiences as it propagates.



```
emf 13.sci (C:\Users\abhis\Documents\emf 13.sci) - SciNotes
File Edit Format Options Window Execute ?
emf 13.sci (C:\Users\abhis\Documents\emf 13.sci) - SciNotes
Untitled 1 x Untitled 2 x emf 13.sci x

1 |clc;
2 |clear;
3 |close;
4 |
5 |//Input values
6 |ur = input("Enter relative permeability (ur):-");
7 |er = input("Enter relative permittivity (er):-");
8 |
9 |//Constants
10 |uo = 4*pi*10^-7;
11 |eo = 8.854*10^-12;
12 |
13 |//Permeability and permittivity
14 |mu = uo * ur;
15 |epsilon = eo * er;
16 |
17 |//Intrinsic impedance
18 |eta = sqrt(mu / epsilon);
19 |
20 |disp("Intrinsic impedance (Ohm) :-", eta);
21 |
22 |//Phase velocity
23 |v = 1 / sqrt(mu * epsilon);
24 |
25 |disp("Phase velocity (m/s) :-", v);
26 |
27 |//Plot
28 |figure();
29 |
30 |x = linspace(0, 10, 50);
31 |
32 |y = eta * exp(0.1 * x);
33 |z = v * exp(-0.1 * x);
34 |
35 |subplot(2, 2, 1);
36 |plot2d3(x, y);
37 |title("Intrinsic Impedance");
```



Enter relative permeability (ur): $5 \cdot 10^{-3}$

Enter relative permittivity (er): $3 \cdot 10^{-3}$

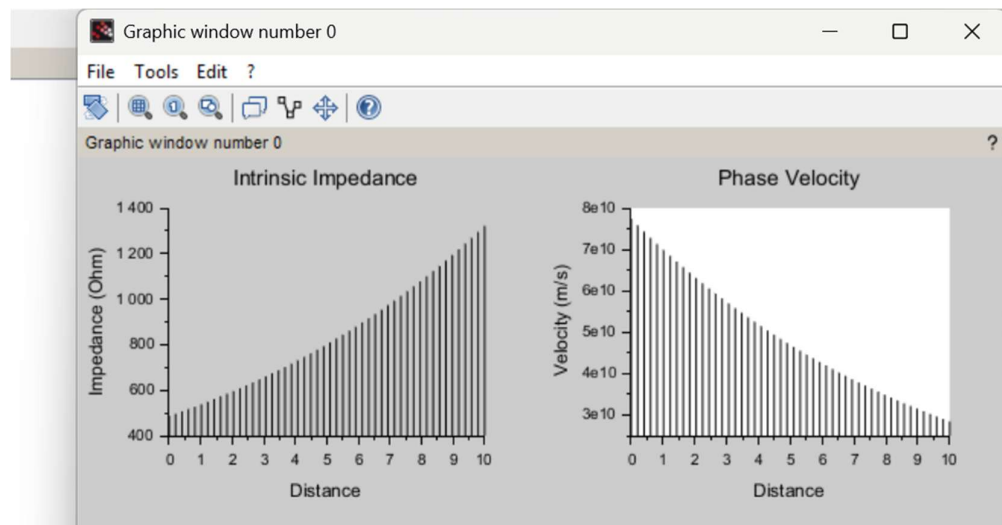
"Intrinsic impedance (Ohm) = "

486.36190

"Phase velocity (m/s) = "

7.741D+10

-->



EXP 13
 $\mu_0 = 4\pi \times 10^{-7}$
 $\epsilon_r = 3 \times 10^{-3}$
 $\epsilon_0 = 8.854 \times 10^{-12}$

Formula

$$\eta = \sqrt{\frac{\mu_0 \mu_r}{\epsilon_0 \epsilon_r}}$$

$$\eta = \sqrt{\frac{(4\pi \times 10^{-7}) (5 \times 10^{-3})}{(8.854 \times 10^{-12}) (3 \times 10^{-3})}}$$

$$\eta = 486.36 \Omega$$

result
 $486.36190 \Omega //$

Formula

$$v = \frac{1}{\sqrt{\mu_0 \mu_r \epsilon_0 \epsilon_r}}$$

$$v = \frac{1}{\sqrt{(4\pi \times 10^{-7}) (5 \times 10^{-3}) (8.854 \times 10^{-12}) (3 \times 10^{-3})}}$$

$$v = 7.74 \times 10^{10} \text{ m/s}$$

result
 $7.74 \times 10^{10} \text{ m/s} //$

Case 1: To determine the intrinsic impedance and phase velocity of electromagnetic waves in free space

(vacuum).

Analytical:

Calculate the Intrinsic Impedance and Phase Velocity of Electromagnetic Waves in Free Space (Vacuum)

```
13.1.sci (C:\Users\abhis\Documents\13.1.sci) - SciNotes
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13.1.sci (C:\Users\abhis\Documents\13.1.sci) - SciNotes
13.1.sci x Untitled 2 x emf 13.sci x

1
2
3
4 //Constants
5 mu0 = 4.*pi.*10^(-7); //Permeability of free space (H/m)
6 epsilon0 = 8.854.*10^(-12); //Permittivity of free space (F/m)
7
8 //Intrinsic Impedance
9 eta0 = sqrt(mu0 / epsilon0);
10
11 //Phase Velocity
12 vp = 1./sqrt(mu0.*epsilon0);
13
14 //Display results
15 disp("=====");
16 disp("...ELECTROMAGNETIC WAVE CALCULATION");
17 disp("=====");
18
19 fprintf("Permeability of free space = %e H/m\n", mu0);
20 fprintf("Permittivity of free space = %e F/m\n", epsilon0);
21 fprintf("Intrinsic Impedance = %2f Ohm\n", eta0);
22 fprintf("Phase Velocity = %2e m/s\n", vp);
23
24 disp("=====");
25
```

```
Scilab 2026.1.0 Console
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Scilab 2026.1.0 Console

"=====
"  ELECTROMAGNETIC WAVE CALCULATION
"=====

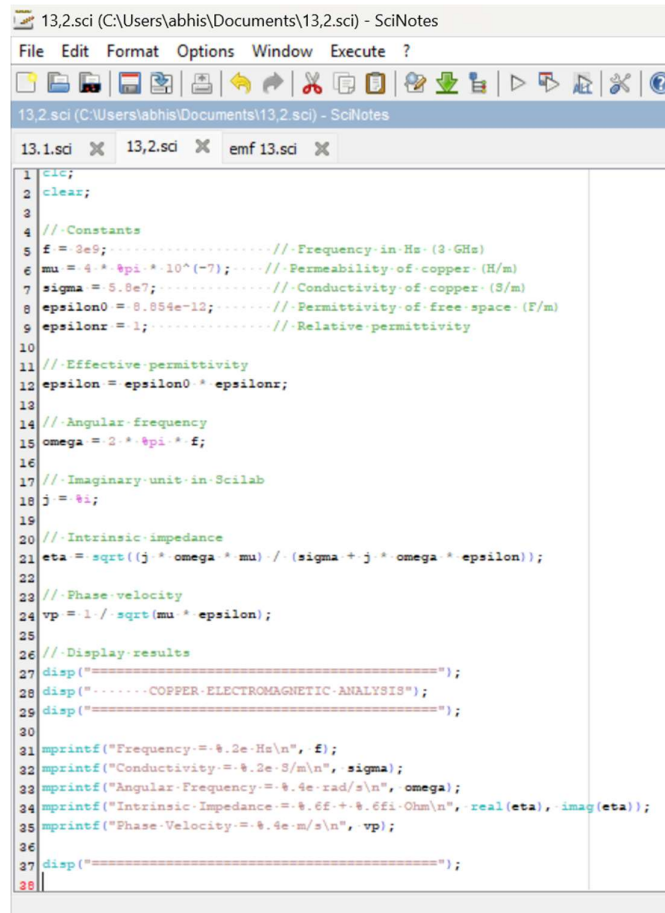
Permeability of free space = 1.256637e-06 H/m
Permittivity of free space = 8.854000e-12 F/m
Intrinsic Impedance = 376.73 Ohm
Phase Velocity = 2.998e+08 m/s

"=====

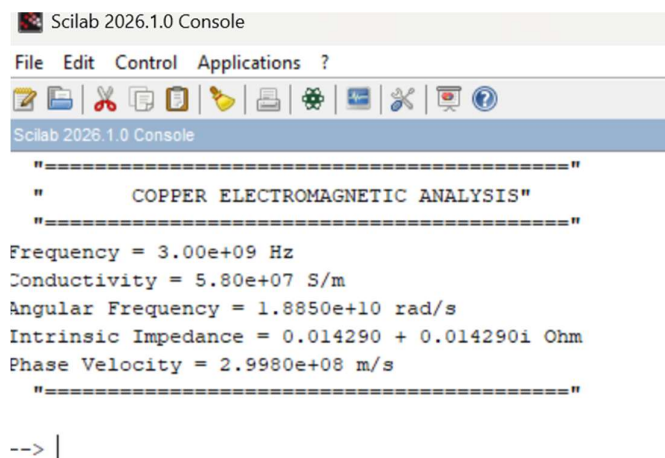
-->
```

Case 2: To determine the intrinsic impedance and phase velocity for a wave propagating through a conductive medium like copper.

Analytical: Calculate the Intrinsic Impedance and Phase Velocity of Electromagnetic Waves in Copper



```
13,2.sci (C:\Users\abhis\Documents\13,2.sci) - SciNotes
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13,2.sci (C:\Users\abhis\Documents\13,2.sci) - SciNotes
13.1.sci x 13,2.sci x emf.13.sci x
1
2 clear;
3
4 // Constants
5 f = 3e9; // Frequency in Hz (3 GHz)
6 mu = 4 * pi * 10^(-7); // Permeability of copper (H/m)
7 sigma = 5.8e7; // Conductivity of copper (S/m)
8 epsilon0 = 8.854e-12; // Permittivity of free space (F/m)
9 epsilonR = 1; // Relative permittivity
10
11 // Effective permittivity
12 epsilon = epsilon0 * epsilonR;
13
14 // Angular frequency
15 omega = 2 * pi * f;
16
17 // Imaginary unit in Scilab
18 j = %i;
19
20 // Intrinsic impedance
21 eta = sqrt((j * omega * mu) / (sigma + j * omega * epsilon));
22
23 // Phase velocity
24 vp = 1 / sqrt(mu * epsilon);
25
26 // Display results
27 disp("=====");
28 disp(".....COPPER ELECTROMAGNETIC ANALYSIS");
29 disp("=====");
30
31 mprintf("Frequency = %.2e-Hz\n", f);
32 mprintf("Conductivity = %.2e-S/m\n", sigma);
33 mprintf("Angular Frequency = %.4e-rad/s\n", omega);
34 mprintf("Intrinsic Impedance = %.6f + %.6fi-Ohm\n", real(eta), imag(eta));
35 mprintf("Phase Velocity = %.4e-m/s\n", vp);
36
37 disp("=====");
38
```



```
Scilab 2026.1.0 Console
File Edit Control Applications ?
Scilab 2026.1.0 Console
"=====
"      COPPER ELECTROMAGNETIC ANALYSIS
"=====
Frequency = 3.00e+09 Hz
Conductivity = 5.80e+07 S/m
Angular Frequency = 1.8850e+10 rad/s
Intrinsic Impedance = 0.014290 + 0.014290i Ohm
Phase Velocity = 2.9980e+08 m/s
"=====

--> |
```

Case 3: To determine and compare the intrinsic impedance and phase velocity of electromagnetic waves propagating through different media: Vacuum, Air, and Water.

Analytical: Determine and compare the intrinsic impedance and phase velocity of electromagnetic waves propagating through different media

```
13.3.sci (C:\Users\abhis\Documents\13.3.sci) - SciNotes
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13.1.sci x 13.2.sci x emf 13.sci x 13.3.sci x

22 //-----
23 mu_water = mu0;
24 epsilon_water = 80 * epsilon0;
25
26 //Function to calculate properties
27 function [eta, vp] = calculate_properties(mu, epsilon)
28     eta = sqrt(mu / epsilon); .....//Intrinsic Impedance (Ohm)
29     vp = 1 / sqrt(mu * epsilon); .....//Phase Velocity (m/s)
30 endfunction
31
32 //Calculate for Vacuum
33 [eta_vacuum, vp_vacuum] = calculate_properties(mu_vacuum, epsilon_vacuum);
34
35 //Calculate for Air
36 [eta_air, vp_air] = calculate_properties(mu_air, epsilon_air);
37
38 //Calculate for Water
39 [eta_water, vp_water] = calculate_properties(mu_water, epsilon_water);
40
41 //Display results
42 disp("=====");
43 disp("... INTRINSIC IMPEDANCE AND PHASE VELOCITY");
44 disp("=====");
45
46 sprintf("Vacuum:\n");
47 sprintf("Intrinsic Impedance = %.2f Ohm\n", eta_vacuum);
48 sprintf("Phase Velocity = %.2e m/s\n", vp_vacuum);
49
50 sprintf("Air:\n");
51 sprintf("Intrinsic Impedance = %.2f Ohm\n", eta_air);
52 sprintf("Phase Velocity = %.2e m/s\n", vp_air);
53
54 sprintf("Water:\n");
55 sprintf("Intrinsic Impedance = %.2f Ohm\n", eta_water);
56 sprintf("Phase Velocity = %.2e m/s\n", vp_water);
57
58 disp("=====");
59
```

```
Scilab 2026.1.0 Console
File Edit Control Applications ?
Scilab 2026.1.0 Console

"=====
" INTRINSIC IMPEDANCE AND PHASE VELOCITY"
"=====

Vacuum:
Intrinsic Impedance = 376.82 Ohm
Phase Velocity = 2.999e+08 m/s

Air:
Intrinsic Impedance = 376.82 Ohm
Phase Velocity = 2.999e+08 m/s

Water:
Intrinsic Impedance = 42.13 Ohm
Phase Velocity = 3.353e+07 m/s
"=====

--> |
```

Result: Thus the determination and plotting of intrinsic and phase velocity is performed and successfully verified in copper.